

BANGLADESH UNIVERSITY OF BUSINESS AND TECHNOLOGY (BUBT)

PROJECT PROPOSAL

RecycleIQ

An AI-Powered Waste Identification, Recycling Guidance, and Community Waste Reporting Web Application

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Submitted By

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1. The Problem

Introduction

Waste segregation and recycling depend on people correctly identifying waste materials and knowing how to dispose of them. In practice, users may be uncertain whether an item belongs in plastic, paper, metal, glass, or organic waste, and local disposal information may be scattered across different sources. Communities also need a simple way to report recurring waste accumulation and view where reports are concentrated.

Problem statements

- People may not be able to identify common waste categories accurately from appearance alone.
- Recycling and disposal guidance can be difficult to find or understand, especially when the user does not know the correct material name.
- Community waste accumulation is often reported through informal channels, making reports difficult to organize, locate, and track in one place.
- Local organizations and project administrators may lack a simple dashboard for reviewing reports and observing basic waste-category and location trends.
- Image-classification predictions can be uncertain or wrong, so the system must communicate confidence and allow users to correct results.

Need for the proposed system

RecycleIQ will combine image-based waste classification, curated disposal guidance, user-submitted geotagged reports, and an administrative dashboard in one web application. The project will be developed as a student prototype, with a defined set of waste categories and a limited pilot dataset. It will not claim to replace professional waste-management services or guarantee that every item is recyclable.

2. Objectives

Project description

RecycleIQ is an AI-integrated web application intended to help users recognize common waste categories and take an informed next step. Users will upload an image, receive a predicted category with a confidence score, read disposal guidance, and optionally submit a location-based waste report. An administrator will review reports and view summary analytics.

Objectives

- Develop a responsive web interface for waste image upload and result display.
- Implement a transfer-learning image classifier for a limited set of common waste categories.
- Provide clear, curated disposal instructions linked to each supported category; an optional language-model layer may simplify the wording but will not be treated as the source of truth.
- Allow users to submit geotagged waste reports with a category, description, image, and status.
- Display submitted reports on an interactive map and provide basic category, date, and status filters.
- Create an administrator dashboard to moderate reports and view summary statistics.

- Evaluate the classifier using a held-out test set and document limitations, errors, and future improvements.

Expected outcome
A deployed, demonstrable web application with a working AI classification flow, waste guidance, report submission, map visualization, administrative review, and basic analytics.

3. Existing System Analysis

3.1 Recycle Coach

Recycle Coach provides waste and recycling information, collection calendars, reminders, educational materials, and a 'What Goes Where' search tool. Its features focus on helping residents follow local waste rules and engage with municipal services [1]. Shortcoming relative to this proposal: its main offering is a municipal information and engagement platform; RecycleIQ's student prototype will emphasize an image-first waste classification workflow integrated with community-submitted, map-based reports. This is a difference in project focus, not a claim that Recycle Coach lacks all image-related features.

3.2 Earth911 / iRecycle

Earth911 offers a recycling search directory that helps users identify recycling options by material and location, and its iRecycle app connects users with recycling opportunities and location information [2][3]. Shortcoming relative to this proposal: the service is strongly oriented toward finding disposal and recycling locations, particularly within its supported geographic coverage. RecycleIQ will focus on image-based category prediction and community waste-report visualization for a small, locally configured pilot area. RecycleIQ will not claim to have a comprehensive or verified recycling-center directory.

3.3 TrashOut

TrashOut enables users to report illegal dumps, view waste locations on a map, find recycling points, and follow report status and environmental activities [4]. Shortcoming relative to this proposal: RecycleIQ's main distinguishing workflow is classifying an uploaded item image and connecting the prediction to disposal guidance before or alongside a community report. TrashOut already covers important community mapping and reporting functions, so RecycleIQ should be presented as a focused educational prototype rather than a replacement for it.

Comparative conclusion
The reviewed systems demonstrate that disposal guidance, recycling-location search, and community reporting are established needs. RecycleIQ's proposed contribution is a compact combination of image classification, category-linked guidance, and a report map in one student-built web application. The project will validate usability and model performance on a limited set of supported categories rather than claim superiority over existing services.

4. Features / Scopes

In-scope features

- User interface: responsive home page, classification page, report form, map, history where applicable, and help/about page.
- Image upload: accept common image formats, validate file type and size, preview the image, and handle invalid input.
- AI classification: classify an image into a defined set of categories such as cardboard, paper, glass, metal, plastic, and organic waste, subject to dataset quality and final model performance.
- Prediction display: show predicted category and confidence score; if confidence is below a chosen threshold, show an uncertain-result message rather than a definitive answer.
- Disposal guidance: display administrator-maintained guidance for each supported category. Guidance will be qualified by local rules and material condition.
- Waste reporting: allow a user to submit a report with category, short description, optional photo, map location or manually entered coordinates, and timestamp.
- Interactive map: display reports with category/status markers and basic filters. Map tiles and geocoding availability will depend on the selected provider's usage policies.
- Admin dashboard: review, approve, reject, or mark reports as resolved; maintain category guidance; view summary counts and category trends.
- Basic analytics: number of detections, category distribution, report statuses, and report counts over time.
- Testing and deployment: test core workflows, document setup, deploy a demonstrable version, and prepare a user guide.

Out of scope

- Automated municipal pickup scheduling or integration with municipal systems.
- Guaranteed identification of mixed, dirty, damaged, or partially obscured waste.
- Verified nationwide recycling-center coverage or live collection availability.
- Predictive waste-volume forecasting requiring a large, reliable historical dataset.
- Native Android/iOS apps, IoT smart bins, payment systems, or a marketplace.
- Claims that the system can determine material composition or recyclability with certainty from an image alone.

Deliverables

- Source code repository and deployment instructions.
- Responsive web application with the agreed MVP features.
- Trained or fine-tuned model and model evaluation summary.
- Database schema and API documentation.
- Test cases, known limitations, user guide, and final demonstration.

5. Development Model

Selected model: Agile / iterative development
An iterative Agile approach will be used because the project combines frontend, backend, computer vision, mapping, and data validation. The team can implement a small end-to-end workflow early, test it, and refine it based on findings instead of waiting until the end to integrate all components.

Reasons
- Requirements can be prioritized into a realistic minimum viable product (MVP).
- AI model performance may require changes to categories, preprocessing, or the user experience.
- Early integration reduces the risk of discovering API, database, or deployment problems in the final week.
- Weekly demonstrations provide checkpoints for team members and the supervisor.
- Out-of-scope features can be deferred without preventing completion of the core workflow.

Working method
Use a prioritized backlog, weekly task planning, Git-based version control, issue tracking, code review for shared modules, and a weekly demonstration. Each week will end with a testable increment.

6. Feasibility Analysis

6.1 Technical feasibility
The project is technically feasible using React, Django REST Framework, PostgreSQL, a pretrained convolutional neural network, and a mapping library. Transfer learning reduces the need to train a large model from scratch [5]. The model will be restricted to a manageable number of classes and evaluated on held-out images. A standard laptop can support web development and small experiments; heavier training can be run in a hosted notebook if needed.

6.2 Economic feasibility
The project can be developed primarily with open-source software. The estimated budget is a planning allowance in Bangladeshi Taka (BDT), not a vendor quotation. Costs can be reduced by using free tiers, a free subdomain, and local development. Paid API usage is optional and will be avoided for core classification.

6.3 Operational feasibility
The application is designed for browser access and a simple user workflow. It can be demonstrated to students, residents, or community users. Local disposal instructions and report moderation will require human review; the application will not independently authorize waste collection or make safety-critical decisions.

8. Requirements

Hardware

- Development computers: at least 8 GB RAM recommended; 16 GB preferred for smoother model experiments.
- Stable internet connection for package installation, source control, deployment, and optional hosted training.
- Optional access to Google Colab or a similar notebook service if local training is slow.

Software and tools

- Operating system: Windows, Linux, or macOS.
- Frontend: HTML, CSS, JavaScript/TypeScript, React, and a UI styling library.
- Backend: Python and Django REST Framework [6].
- AI/ML: TensorFlow/Keras or a compatible pretrained image-classification model; scikit-learn for evaluation metrics.
- Database: PostgreSQL for deployment; SQLite may be used during initial prototyping.
- Maps: Leaflet with a suitable tile provider [7].
- Version control and collaboration: Git and GitHub.
- Design and testing: Figma or equivalent, browser developer tools, API client, and automated/manual test cases.
- Deployment: a frontend host and a Python-compatible backend/database host; free-tier availability and limits must be checked at deployment time.

Personnel

Five student developers with assigned responsibilities: project coordination and frontend, backend/database, and AI/model evaluation. All members will participate in integration, testing, documentation, and demonstration.

9. Activity and Time Schedule



Effort assumptions

The estimate is 240 person-hours in total, approximately 6 hours per week per team member for a five-person team over eight weeks.

10. Budget

Estimated student-project budget (BDT)

Project item	Quantity / basis	Estimated unit cost (BDT)	Estimated total (BDT)
Software licenses	Open-source tools	0	0
Development machines	Existing student laptops	0 incremental	0
Model training	Local CPU / available free notebook tier	0 assumed	0
Domain name (optional)	1 year	1,500	1,500
Hosting / database allowance	8 weeks; free tier preferred	3,000	3,000
Optional LLM/API allowance	Limited testing only	2,000	2,000
Internet / communication allowance	Project period	2,000	2,000
Contingency	Unexpected small costs	1,500	1,500
Total Estimated Budget			10,000

Budget notes

The estimated total is BDT 10,000 and is not a supplier quotation. It assumes the team already has computers and can use open-source software. The optional LLM/API allowance can be removed if all guidance is curated and stored locally. Actual hosting, domain, and API charges depend on the provider, usage, exchange rate, and current pricing. The project can target a lower cost by using free tiers and not purchasing a domain.

11. Development Team & Stakeholders

Team structure

Student ID	Name	Designation	Responsibilities
20234103118	Vasha Quddus	Team Leader / AI/ML Lead	Prepare dataset, train/evaluate image classifier, integrate inference, report model metrics and support testing.
20234103201	Md Al-hanullah Sajib	Co-Leader / Frontend Lead	Design database and Django REST APIs, implement report workflow, admin moderation, validation and deployment support.
20234103116	Md Nafijur Rahaman	Backend Lead	Design the database and Django REST APIs, implement the report workflow, admin moderation and validation, and support deployment.

20234103115	Shad Mohammad	UI/UX and Analytics Developer	Produce wireframes and the design system. Build the admin dashboard, summary charts, category guidance pages and map filters. Handle responsive layout and usability fixes.
20234103205	Arifa Islam	QA, DevOps and Documentation Lead	Write test plans and run functional, API, model and usability tests. Set up the repository, CI and deployment, and write the README, user guide and final evaluation report.

Shared responsibilities

All members will review code, test integrated features, maintain documentation, participate in weekly planning, and prepare the final demonstration. Tasks will be assigned so that no individual task requires more than two people as specified in the provided proposal format.

Stakeholders and users

- Project supervisor and department: review scope, progress, deliverables, and academic quality.
- General users/students/residents: classify waste, view guidance, and submit reports.
- Administrator/moderator: review submitted reports, update guidance, and maintain data quality.
- Potential community or environmental organizations: possible future users or reviewers; no partnership is assumed in this proposal.

12. Limitations & Future Works

Current limitations

- Classification quality depends on the dataset, image quality, lighting, background, and whether the image contains one clearly visible item.
- The initial model will cover only a limited number of categories and may confuse visually similar materials.
- A predicted category does not establish the exact material composition, contamination level, or local recyclability of an item.
- Disposal guidance must be checked against local rules; the project will not guarantee a collection service or recycling facility.
- Community map accuracy depends on user-provided locations, report quality, and moderation.
- Real-time waste forecasting and impact estimates are not included because reliable longitudinal data may not be available.

Future works

- Expand the dataset with locally collected and reviewed images, including Bangla-labeled examples.
- Add multi-object detection or segmentation for images containing several waste items.
- Introduce a reviewed, location-aware disposal knowledge base for more areas.
- Integrate verified recycling centers and collection partners where data access and permission are available.

- Add Bangla voice guidance and accessibility improvements.
- Use duplicate detection and report prioritization to support moderators.
- Explore waste-volume trend analysis after enough validated reports have accumulated.
- Conduct a larger usability study and evaluate model fairness across image conditions.

13. References

Works cited

- [1] Recycle Coach. “Build Better Habits & Get Wise About Waste.” Recycle Coach Solutions, https://www.recyclecoach.com/. Accessed 17 Oct. 2026.
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